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MILITARY SPECIFICATION

RELAY ASSEMBLY, ARMING, MAGNETIC RYU-7(*)/A24 AND RACK, ELECTRICAL EQUIPMENT MTU-6(*)/A24 (U)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be an automatic control unit of the Special Weapons Monitor and Control System for the DSN-3 Drone Anti-Submarine Helicopter and shall, in response to d-c command signals, provide the necessary switching to monitor and control power to the DSN-3 armament.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Items	Type Designation	Appl. Para.
Relay Assembly, Arming, Magnetic	RYU-7(*)/A24	3.4.1
Rack, Electrical Equipment	MTU-6(*)/A24	3.4.2

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes, Choice and application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

DOWNGRADED AT 3 YEAR INTERVALS;
DECLASSIFIED AFTER 12 YEARS.
DOD DIR 5200.10

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MIL-T-5422	Environmental Testing, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MS-25245	Connections, Input Power Airborne Electronic Equipment; 28-volt DC, 115-Volt Single Phase AC, and 115/200-Volt three phase AC or combination

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5)

(3) When previously produced models of this equipment used repairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors, and diodes shall be chosen and

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applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 8 pounds.

3.2.3 Operating Life -

3.2.3.1 Mean Operating Time Between Failures - The equipment shall have a minimum of 20,000 cycles* of operation between failure. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

*NOTE: A complete cycle of operation for the equipment is defined as the unit being energized, and operated from the off condition through safe to arm, then reverse the procedure from arm through safe to off, and de-energize the equipment.

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Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein.

3.2.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and III of Specification MIL-E-5400.

3.2.11 Warm-Up Time - The equipment shall be capable of providing the performance characteristics specified herein without the necessity of a warm-up period.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. Operation under emergency conditions shall be with a a minimum of 18 volts dc in lieu of that specified in MIL-STD-704.

(1) DC Power 28-Volt System, Category "B",

Steady-state d-c voltage	18-30
Current: The current shall be as specified for each of the respective input conditions;	
Off (no input signal)	0.04 Amps
Command safe signal on J102	0.1 Amp
Additional power required during command safe on J101	0.6 to 6.0 Amps
Command arm signal on J102	0.6 Amp
Additional power required during command arm	0.6 to 6.0 amps, plus a 45 amp surge.
Command off signal on J102	0.1 Amp
Additional power required during command off on J101	0.04 to 6.0 Amps

3.2.12.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Relay Assembly, Arming, Magnetic RYU-7/A24 - The relay assembly

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Arming, Magnetic RYU-7/A24 shall when provided with the specified power (refer to ~~paragraph~~ 3.2.12), provide remote switching necessary to monitor and control the power to the DSN-3 arming circuits. All switching shall occur in response to command type signals from a control telemetering system.

3.3.3 Input-output Performance Characteristics - The relay assembly shall be capable of providing the performance requirements specified below when provided with the specified command signal. Outputs or extraneous voltages at the connectors, connector pins, etc. shall not be permitted unless they are specified for the particular condition or input.

3.3.3.1 Command Off Signal - The command off signal shall be an 18-to-30-volt, 100 milliampere d-c input. This signal shall disable the relay assembly and make it incapable of responding to other command inputs. The outputs from and the continuity requirements of the relay assembly shall be the same as those provided at no signal input and shall be as follows:

a. The OFF indicator lamp shall be illuminated.

b. The input power to pins B and C of J101 shall be applied through the relay assembly to J103-B.

c. Continuity shall be provided from pins K to E and pins J to P of J104.

3.3.3.2 Command Safe Signal - The command safe signal shall be an 18-to-30-volt, 100 milliampere input. An additional 18-to-30-volt input to the control winding of K106 shall be required to place the relay assembly in the safe condition; once in the safe condition, the assembly shall continue to function as described below even after the command safe input is removed and shall be capable of being placed in the command arm or command off condition by application of the respective command signal:

a. The SAFE indicator lamp shall be illuminated.

b. The input power to pins B and C of J101 shall be coupled through the relay assembly to pins J104-O, J104-I, and J103-D.

c. Continuity shall be provided from pins K to E and pins J to P of J104.

3.3.3.3 Command Arming Signal - The command arming signal shall be an 18-to-30-volt, 600 millampere input. The relay assembly shall be capable of being placed in the command arm condition from the command safe condition only and shall remain in the command arm condition only while a command arm signal is applied. The relay assembly shall provide the following performance characteristics:

a. The input power to pins B and C of J101 shall be coupled through the relay assembly to pins J104-O and J104-I. These circuits shall be capable of accomodating the following current loads:

(1) Maximum surge - A maximum surge rising from 0 to 45 amperes with a rise time of 10 milliseconds, and a total time duration of 20 milliseconds including rise time.

(2) Steady-state - A steady state current of 6 amperes.

b. Continuity shall be provided from pins H to P and F to E of J104.

3.3.4 Monitor Arming Circuit - Continuity shall be provided between J104-L and J103-F

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at all times. An Arming indicator lamp shall be provided in this circuit to indicate the presence of an 18-to-30-volt input on either J104-L or J103-F.

3.3.5 Multiple Command Signals - The design of the relay assembly shall be such that if two or more command signals are applied to the equipment at the same time, the relay assembly will switch automatically to the command safe or command off condition depending upon the combination of inputs and the initial condition of the assembly.

3.3.6 Power Switch - The relay assembly shall be equipped with a manually operated power switch to interrupt power to the relay circuitry.

3.3.7 Fuse Test Switch and Indicator Lamp - A test switch and indicator lamp shall be provided to permit testing each of the power input fuses.

3.4 Detail Requirements -

3.4.1 Relay Assembly, Arming, Magnetic RYU-7/A24 - The relay assembly shall be designed to meet the following requirements:

3.4.1.1 Form Factor - The maximum allowable dimensions excluding shock mount shall not exceed 10-11/32 wide by 5-29/32 deep by 6-11/16 inches high.

3.4.1.2 Weight - The weight of relay assembly shall not exceed 8 pounds.

3.4.1.3 Contents of Unit - The relay assembly shall contain all the circuits necessary to provide the performance characteristics specified herein.

3.4.1.4 Controls - That portion of the front panel containing the controls shall be constructed as a recessed compartment with a hinged lid. The lid shall be equippped with a suitable gasket to seal off the recessed portion and prevent moisture from seeping into the controls. The hinge of the lid shall anchor at the bottom permitting the lid to swing downward in order to gain free access to the controls. The fasteners which lock the lid in a closed position shall be of a type which can be easily operated with a gloved hand. The recessed portion of the front panel of the relay assembly shall provide the following controls:

a. Switch S101-

b. Switch S102-

3.4.1.5 Indication (Visual) - The front panel of the relay assembly shall provide the following visual indicators: Press to test features shall be incorporated in each of the indicator lamp circuits:

a. DS101 (Green) OFF

b. DS102 (Yellow) SAFE

c. DS103 (Red) ARM

d. DS104 (White) Fuse test lamp

3.4.1.6 Power Input Fuses - The power input circuits pins B and C (E ground) of J101 (MS3102R-14S-5P) shall be protected by 15 amp fuses. Access to these fuses shall be provided on the front panel. The front panel shall also provide an additional fuse as a spare. The fuses are as follows:

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a. Fuse F101 15 Amp

b. Fuse F102 15 Amp

3.4.1.7 Relays - The circuitry of the relay assembly shall contain the following relays:

a. Relay K101 (Command off relay)

b. Relay K102 (Command safe relay)

c. Relay K103 (Auxiliary safe relay)

d. Relay K104 (Latching ON-OFF relay)

e. Relay K105 (Safe relay)

f. Relay K106 (Monitor safe relay)

g. Relay K107 (Command arm relay No. 1.)

h. Relay K108 (Command arm relay No. 2.)

3.4.1.8 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J101	MS3102R-14S-5P	Power Connector
J102	MS3102R-18-1P	Control Telemeter Connector
J103	MS3102R-18-8S	Monitor Telemeter Connector
J104	MS3102R-24-7S	Weapons Connector
J105	MS3102R-14S-5S	Weapons Connector

3.4.2 Rack, Electrical Equipment MTU-6/A24 - The rack shall meet the following requirements.

3.4.2.1 Function - Rack, Electrical Equipment MTU-6/A24 shall be used for the special purpose of providing a shock mount for Relay Assembly, Arming, Magnetic RYU-7/A24.

3.4.2.2 Form Factor and Mounting - The rack shall mount on a flat horizontal surface. The maximum allowable dimensions shall be 10-23/32 by 7-3/4 by 2-1/8 inches.

3.4.2.3 Weight - The weight of the rack shall not exceed 1 pound 6 ounces.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) *Preproduction Tests*

(2) Production Sample Tests

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(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requrested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.3.3.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

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4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two-3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipment for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

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(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)

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Humidity	Room ambient
D. C. voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to

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comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*) when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

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